common bile duct and enter into the duodenum. The cholesterol content of bile can crystallize and grow in size, leading to formation of gallstones. Sometimes, the stones might block the common bile duct and cause obstructive jaundice, in which case, the gallbladder is removed.

2.1.2.5 Pancreas

The pancreas lies deep in the abdominal cavity, behind the stomach in the upper left abdomen resting on the posterior abdominal wall. The broad portion of the pancreas is called as the head and it fills the loop formed by the duodenum. The narrow portion is called the tail. The pancreas has both endocrine (as it secretes glucagon and insulin into the blood stream) and exocrine function (by way of secreting pancreatic juice). Pancreatic acinar cells produce pancreatic juice, which is secreted into tiny tubes that merge forming a single pancreatic duct that extends the length of the pancreas. This pancreatic duct joins the common bile duct to form the hepatopancreatic duct Ampulla of Vater that is guarded by the Sphincter of Oddi.

Glands of Lieberkuhn

These are tubular glands that lie between the villi of the inner surface of the small intestine.

2.1.2.6 Brunner's glands

The submucosa of the duodenum above the Oddi of Sphincter contains duodenal glands, also called Brunner's glands, which secrete alkaline mucus into the intestinal lumen. The functions of the mucus are:

- To neutralize the acid in the chyme
- To provide an alkaline pH enabling the intestinal enzymes to act
- To protect the intestinal walls

The secretions of the Brunner's gland and Glands of Lieberkuhn together contribute to intestinal juice. About 1 -2 liters of intestinal juice is secreted every day.

2.2. DIGESTION

GASTROINTESTINAL TRACT	SECRETORY GLANDS
Oral cavity	Salivary glands
Pharynx	Gastric glands
Oesophagus	Liver and Gall bladder
Stomach	Pancreas
Small Intestine	Duodenal glands (Brunner's glands)
Large Intestine	
Rectum	
Anus	

XII U2 Digestion.indd 34 4/8/2020 3:05:27 PM